SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I AGNEESHWARAN B (192524508) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

AGNEESHWARAN B

Annexure A

Scenario Based Assignment

1. **Scenario** : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic iii. Analyze the impact of heuristic accuracy on both algorithms iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering realtime constraints, optimality, and safety

i. Greedy Best-First Search (GBFS)

Greedy Best-First Search selects the path with the **lowest heuristic value (straight-line distance)** without considering the total travel cost.

Behavior:

- Chooses the route that appears closest to the destination.
- Ignores movement cost due to weather or terrain.
- Quickly reaches a decision.

Strengths

- Very fast.
- Low memory usage.
- Suitable for emergency situations requiring immediate decisions.

Weaknesses

- May choose unsafe or blocked routes.
- Does not guarantee the shortest or safest path.
- Performs poorly when heuristic information is inaccurate.

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ii. A* Search

A* Search evaluates paths using:

f(n) = g(n) + h(n)

Where:

- **g(n)** = Actual cost from start to current node.
- **h(n)** = Estimated remaining distance (heuristic).

Behavior

- Considers both travel cost and estimated distance.
- Avoids expensive terrain.
- Finds an optimal path when the heuristic is admissible.

Advantages

- Produces optimal solutions.
- Better safety.
- Adapts to varying movement costs.

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iii. Impact of Heuristic Accuracy

Greedy Best-First Search	A* Search
Highly dependent on heuristic accuracy	Less dependent because actual cost is included
Poor heuristic gives poor routes	Still finds near-optimal solution
Fast but risky	Slightly slower but reliable

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iv. Effect of Dynamic Changes

- Blocked roads require replanning.
- GBFS may repeatedly choose blocked paths.
- A* updates path costs and searches for alternative routes.
- Frequent updates increase computation but improve safety.

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v. Recommended Algorithm

Recommended: *A Search**

Justification

- Balances speed and optimality.
- Considers terrain and weather costs.
- Produces safer routes.
- Handles dynamic environments more effectively.

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations
ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
iv. Recommend the best approach and justify your choice based on scalability and adaptability

Problem Formulation

Objective

Minimize:

- Vehicle waiting time
- Fuel consumption
- Traffic congestion

Variables

- Signal timing
- Green-light duration
- Signal sequence

Constraints

- Traffic volume
- Emergency vehicles
- Pedestrian crossings

Traditional search becomes inefficient because the number of signal combinations is extremely large.

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i. Hill Climbing

Hill Climbing continuously selects the neighboring solution with the best improvement.

Advantages

- Simple
- Fast
- Low memory

Limitations

- Gets trapped in local maxima.
- Cannot escape plateaus.
- Sensitive to initial solution.

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ii. Local Maxima, Plateaus and Ridges

Local Maximum

A solution better than nearby solutions but not globally optimal.

Example:
Traffic improves in one junction while overall congestion remains high.

Plateau

Many neighboring solutions have equal quality.

Example:
Changing signal timing produces no improvement.

Ridge

Optimal solution requires multiple coordinated changes.

Example:
Changing one signal alone is ineffective.

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iii. Simulated Annealing and Genetic Algorithm

Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.
- Gradually becomes more selective.

Genetic Algorithm

- Uses selection, crossover and mutation.
- Explores multiple solutions simultaneously.
- Finds high-quality solutions for complex optimization.

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iii. Recommended Approach

Genetic Algorithm

Reason

- Highly scalable.
- Adapts to changing traffic.
- Finds near-optimal signal timings.
- Suitable for large smart-city environments.

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search ii.

Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model iv. Compare

online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

i. Why Online Search?

The rover has:

No complete map.

Unknown terrain.

Limited sensing capability.

Therefore, decisions must be made while exploring.

Offline search cannot be used because the complete environment is unavailable.

ii. Challenges

Partial Observability

Hidden rocks

Unknown craters

Limited sensor range

Dynamic Environment

Dust storms

Sensor errors

Newly discovered obstacles

iii. Internal Model

The rover:

Stores explored locations.

Marks obstacles.

Updates safe paths.

Learns from sensor observations.

Its internal map continuously improves during exploration.

iv. Online vs Other Search

Online Search	Uninformed Search	Informed Search
Learns while moving	Requires complete graph	Uses heuristic
Suitable for unknown areas	Poor efficiency	Better when map exists
Highly adaptive	Not adaptive	Semi-adaptive

v. Recommended Approach

Online Search Agent

Reason

Works in unknown environments.

Updates knowledge continuously.

Ensures safer navigation.

Adapts to uncertainty.

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation ii. Explain how backtracking search works in this scenario iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency iv. Analyze real-world challenges and justify the best strategy for scalability

CSP Formulation

Variables

- Courses
- Time slots
- Examination halls

Domains

Available:

- Dates
- Time slots
- Halls

Constraints

- No student has overlapping exams.
- Hall capacity must not exceed limit.
- Faculty must be available.

- Subject precedence rules.
- Special accommodation requirements.

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i. Variables, Domains and Constraints

Variables represent exams.

Domains represent available scheduling options.

Constraints ensure all university rules are satisfied.

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ii. Backtracking Search

1. Assign first exam.
2. Check constraints.
3. If conflict occurs, backtrack.
4. Try another assignment.
5. Continue until all exams are scheduled.

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iii. Forward Checking

Forward checking:

- Removes invalid future assignments.
- Detects conflicts early.
- Reduces unnecessary searching.
- Improves efficiency.

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iv. Real-World Challenges

- Large number of students.
- Hall shortages.
- Faculty availability.
- Last-minute changes.

Recommended Strategy

Backtracking + Forward Checking + Constraint Propagation

This combination efficiently solves large timetable problems.

5. Scenario: Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning iv. Design an evaluation function and justify the factors considered

Recommend a strategy for balancing optimality and real-time performance

i. Minimax Algorithm

Minimax assumes:

- AI tries to maximize score.
- Opponent tries to minimize AI's score.

The AI evaluates all possible moves before selecting the best one.

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ii. Computational Challenges

- Huge branching factor.
- Large search tree.
- High memory usage.
- Slow decision making in real time.

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iii. Alpha-Beta Pruning

Alpha-Beta Pruning eliminates branches that cannot influence the final decision.

Advantages

- Reduces search space.
- Speeds up decision making.
- Produces the same optimal result as Minimax.
- Enables deeper search within the same time.

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iv. Evaluation Function

Factors considered:

- Health points
- Available resources
- Army strength
- Territory control
- Defensive position
- Distance from enemy
- Probability of winning

These factors help estimate the quality of each game state.

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v. Recommended Strategy
Minimax with Alpha-Beta Pruning
Justification

- Maintains optimal decisions.
- Significantly reduces computation.
- Suitable for real-time strategy games.
- Balances accuracy and speed.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand